

Prahanya Sriram

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EDUCATION

Carnegie Mellon University (CMU) | School of Computer Science Pittsburgh, PA

Master of Software Engineering (CGPA 3.88/4.00) Dec. 2025

Relevant Courses: Distributed Systems, Software Architecture, DevOps, Foundations of Computational Data Science, Generative AI, Quality Assurance, Requirements Engineering.

College of Engineering Anna University Chennai, India

Bachelor of Engineering in Computer Science (CGPA 9.2/10.00) Apr. 2020

WORK EXPERIENCE

EParts Services Feb. 2026 - Present

DevOps Engineer

- Designed Bicep-based Infrastructure as Code architecture for Azure Container Apps, App Services, and Service Bus, enabling ~95% deployment automation and reducing manual provisioning.
- Implemented production observability with Datadog APM and AI-assisted incident support, accelerating root-cause analysis and driving team adoption of optimized operational outcomes.
- Spearheaded the development of org-wide feature flagging (flags, targeting, staged rollouts, kill switches) integrated into .NET + Vue.js application, enabling safe continuous delivery and faster incident mitigation without full rollbacks.
- Implemented Cloudflare Web Application Firewall (WAF) and bot mitigation policies to block malicious traffic at the edge, improving application security, reducing backend load, and protecting production services from automated abuse.
- Automated production deployment rollback workflows using Bicep, reducing recovery time by 50% and improving deployment reliability during production incidents.
- Implemented zero-downtime Elasticsearch re-indexing for 5M+ documents using blue/green alias swapping on Azure Durable Functions, eliminating partial data failures during full sync.
- Orchestrated an automated purchase-order export pipeline with ERP-compliant field mapping and validation, eliminating manual formatting effort and improving downstream data reliability.

BNY Mellon Jul. 2020 - May 2024

Software Engineer II

- Redesigned a production-grade file upload service (Java, Spring Boot, Angular), improving fault tolerance and reducing event-driven failures by 45% and tracked the progress of file upload.
- Built a Python validation module to standardize JSON, XML, DAT file ingestion, enforcing schema consistency and improving backend reliability by 26%.
- Designed and delivered 5+ RESTful APIs (Java/Spring Boot/MySQL) meeting less than 120 s SLA targets, enabling resilient client-server communication and data retrieval across enterprise systems.
- Diagnosed and resolved critical production incidents (memory leaks, integration failures) using Splunk and AppDynamics, strengthening system uptime and reliability.
- Automated GitLab CI/CD pipelines with security scans and expanded test coverage by 40% using JUnit/Mockito, cutting deployment time by 22% and regression defects by 35%.

PROJECTS

Unified Data Management Platform (Capstone Project), Carnegie Mellon University Jan. 2025 - Dec. 2025

- Designed a resilient multi-tenant Snowflake warehouse integrated with real-time Kafka CDC pipelines on Kubernetes, cutting SQL server load by 30% and enhancing scalability.
- Implemented a Medallion data architecture to organize raw, curated, and consumer-ready data layers, enabling efficient BI dashboards, analytics, and MLOps pipelines.

- Automated cloud infrastructure provisioning using Infrastructure as Code (IaC) with Terraform and Helm to create a reproducible environment and streamline onboarding of new tenants.

Custom Transport Layer Protocol, Carnegie Mellon University Sep. 2025 - Oct. 2025

- Engineered a reliable client-server messaging protocol over UDP in GO leveraging channels and goroutines to ensure exactly-once, in-order delivery and integrity.
- Built connection management, sliding window retransmission, and exponential backoff mechanisms with heartbeat-based failure detection to maintain robust communication under packet loss.

Multi-Object Cardinality Preservation in Stable Diffusion, Carnegie Mellon University Feb. 2025 - May 2025

- Optimized Stable Diffusion v1.5 with slot-guided and bounded attention mechanisms to advance object separation and ensure each entity in a prompt (Example: "3 cats and a dog") is distinctly represented without semantic leakage.
- Trained and validated models on multi-object cardinality prompts, achieving a CLIP score of 0.32 (+30%), precision of 0.93, and F1 score of 0.75, setting new state-of-the-art benchmarks in number sensitive image generation.
- Achievement: Best Student Paper Award.

SKILLS

Languages: C#, Java, Go, Python, SQL

Backend & Frameworks: .NET, ASP.NET Core, Spring Boot, REST APIs, Microservices, Vue.js

Cloud & Infrastructure: Microsoft Azure (Container Apps, App Services, Service Bus), Docker, Kubernetes, Bicep, Terraform

CI/CD & Observability: GitLab CI/CD, GitHub, Datadog (APM, Error Tracking, SLOs), Splunk, AppDynamics

Data Platforms: Kafka, Snowflake, Debezium, MySQL, Elasticsearch